Red Hat Summit

Mitigating AI's New Risk Frontier: A Practical Guide to Unifying Enterprise AI Security and Safety

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Agenda

- Importance of AI Security and Safety
- Security vs Safety
- Managing Safety Risks
- Mapping Security Risks
- Enterprise Security
- Red Hat AI Approach

Google AI flaw allows indirect prompt injection
siliconANGLE Sep 30, 2025

AgentForce AI prompt injection vulnerability can expose sensitive CRM data
OECD Incidents Sep 25, 2025

An AI-powered coding tool wiped out a software company’s database
Fortune July 23, 2025

ShadowLeak Zero-Click Flaw exfiltrates gmail data using ChatGPT Deep Research Agent
radware Sep 18, 2025

GitLab Duo AI Vulnerability Exposes Source Code and Confidential Data
OECD Incidents May 21, 2025

Ultralytics AI model Compromised: Cryptocurrency Miner Found in PyPI packages
TechTarget Dec 06, 2024

Why it matters?

- **Accelerating** AI deployments across enterprise
- **Increased** attack surface: private data, systems, and AI models become targets
- **Success** requires AI deployments to be secure, compliant, and trustworthy

AI security vs. AI safety

AI Security

Protects confidentiality, integrity and availability of AI systems

Examples:

- Broken authentication
- Prompt injection

Risk: data breach

AI Safety

Ensures AI behaves as intended, aligns with ethics, policies

Examples:

- Bias
- Hallucinations

Risk: loss of trust, ethical damage

Managing safety risks

- **Safety focus** throughout pre-training, fine-tuning and alignment
- **Continuous evaluation** & benchmarking of safety outcomes
- **LLM evaluation harness** to measure likelihood of harmful outputs

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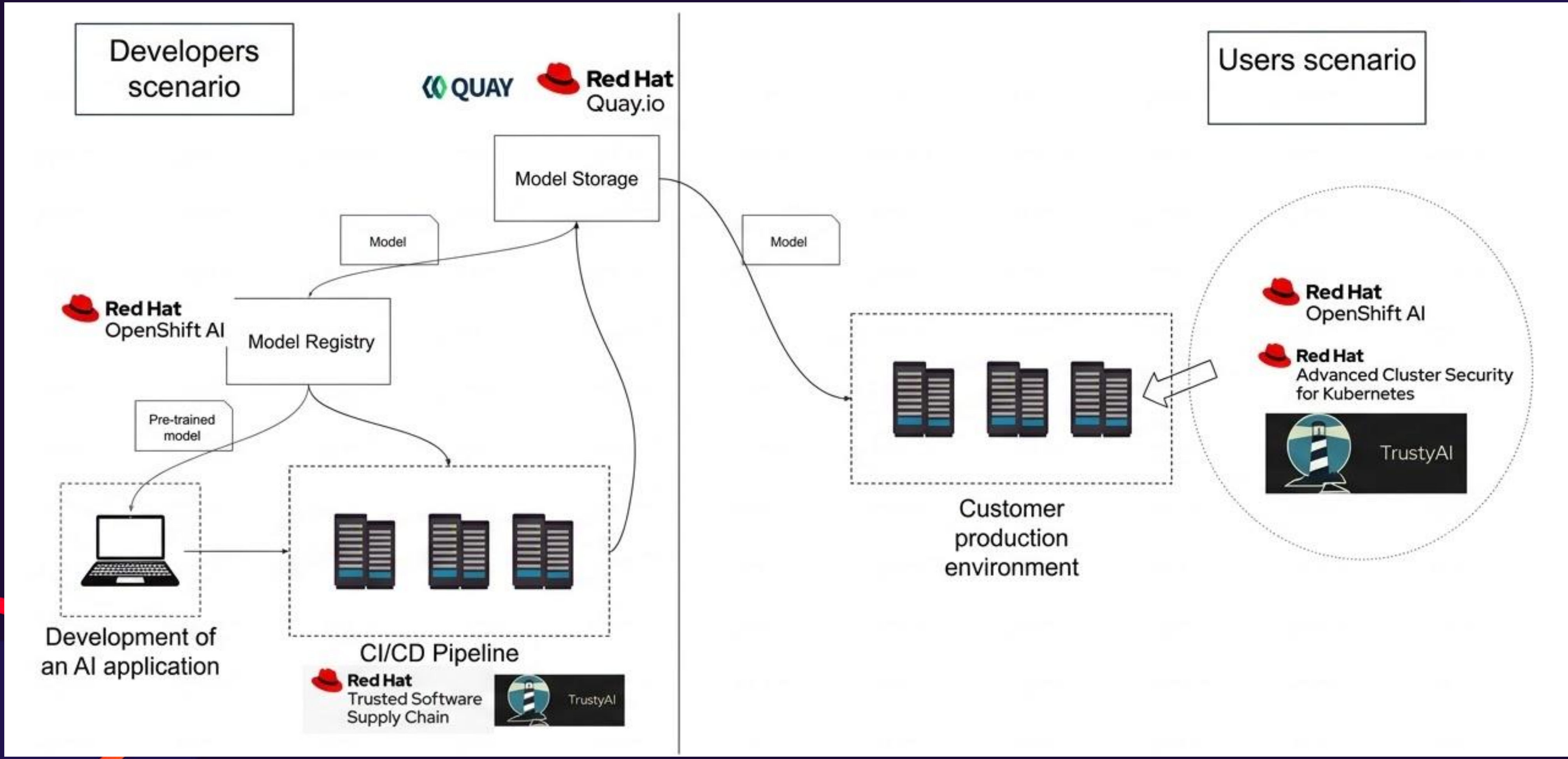
Red Hat OpenShift AI includes TrustyAI

- Evaluation with lm-evaluation-harness
- Guardrails

AI model scenarios



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Mapping security risks

- What can go wrong?
- MITRE ATLAS™, NIST AI RMF, and OWASP Top 10 for LLMs as sources of threats.

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MITRE ATLAS

Reconnaissance&
8 techniques
- Active Scanning&
- Gather RAG-Indexed Targets
- Gather Victim Identity Information&
- Search Application Repositories
- Search Open AI Vulnerability Analysis
- Search Open Technical Databases&
- Search Open Websites/Domains&
- Search Victim-Owned Websites&

Resource Development&
13 techniques
- Acquire Infrastructure
- Acquire Public AI Artifacts
- Develop Capabilities&
- Establish Accounts&
- LLM Prompt Crafting
- Obtain Capabilities&
- Poison Training Data
- Publish Hallucinated Entities
- Publish Poisoned AI Agent Tool
- Publish Poisoned Datasets
- Publish Poisoned Models
- Retrieval Content Crafting
- Stage Capabilities&

Initial Access&
7 techniques
- AI Supply Chain Compromise
- Drive-by Compromise&
- Evade AI Model
- Exploit Public-Facing Application&
- Phishing&
- Prompt Infiltration via Public-Facing Application
- Valid Accounts&

AI Model Access
4 techniques
- AI Model Inference API Access
- AI-Enabled Product or Service
- Full AI Model Access
- Physical Environment Access

Execution&
6 techniques
- AI Agent Clickbait
- AI Agent Tool Invocation
- Command and Scripting Interpreter&
- Deploy AI Agent
- LLM Prompt Injection
- User Execution&

Persistence&
8 techniques
- AI Agent Context Poisoning
- AI Agent Tool Data Poisoning
- LLM Prompt Self-Replication
- Manipulate AI Model
- Modify AI Agent Configuration
- Poison Training Data
- Prompt Infiltration via Public-Facing Application
- RAG Poisoning

Privilege Escalation&
4 techniques
- AI Agent Tool Invocation
- Escape to Host&
- LLM Jailbreak
- Valid Accounts&

Defense Evasion&
13 techniques
- Corrupt AI Model
- Delay Execution of LLM Instructions
- Evade AI Model
- Exploitation for Defense Evasion&
- False RAG Entry Injection
- Impersonation&
- LLM Jailbreak
- LLM Prompt Obfuscation
- LLM Trusted Output Components Manipulation
- Manipulate User LLM Chat History
- Masquerading&
- Modify AI Agent Configuration
- Virtualization/Sandbox Evasion&

Credential Access&
6 techniques
- AI Agent Tool Credential Harvesting
- Credentials from AI Agent Configuration
- Exploitation for Credential Access&
- OS Credential Dumping&
- RAG Credential Harvesting
- Unsecured Credentials&

Discovery&
9 techniques
- Cloud Service Discovery&
- Discover AI Agent Configuration
- Discover AI Artifacts
- Discover AI Model Family
- Discover AI Model Ontology
- Discover AI Model Outputs
- Discover LLM Hallucinations
- Discover LLM System Information
- Process Discovery&

Lateral Movement&
2 techniques
- Phishing&
- Use Alternate Authentication Material&

Collection&
4 techniques
- AI Artifact Collection
- Data from AI Services
- Data from Information Repositories&
- Data from Local System&

AI Attack Staging
6 techniques
- Craft Adversarial Data
- Create Proxy AI Model
- Generate Deepfakes
- Generate Malicious Commands
- Manipulate AI Model
- Verify Attack

Command and Control&
3 techniques
- AI Agent
- AI Service API
- Reverse Shell

Exfiltration&
6 techniques
- Exfiltration via AI Agent Tool Invocation
- Exfiltration via AI Inference API
- Exfiltration via Cyber Means
- Extract LLM System Prompt
- LLM Data Leakage
- LLM Response Rendering

Impact&
8 techniques
- Cost Harvesting
- Data Destruction via AI Agent Tool Invocation
- Denial of AI Service
- Erode AI Model Integrity
- Erode Dataset Integrity
- Evade AI Model
- External Harms
- Spamming AI System with Chaff Data

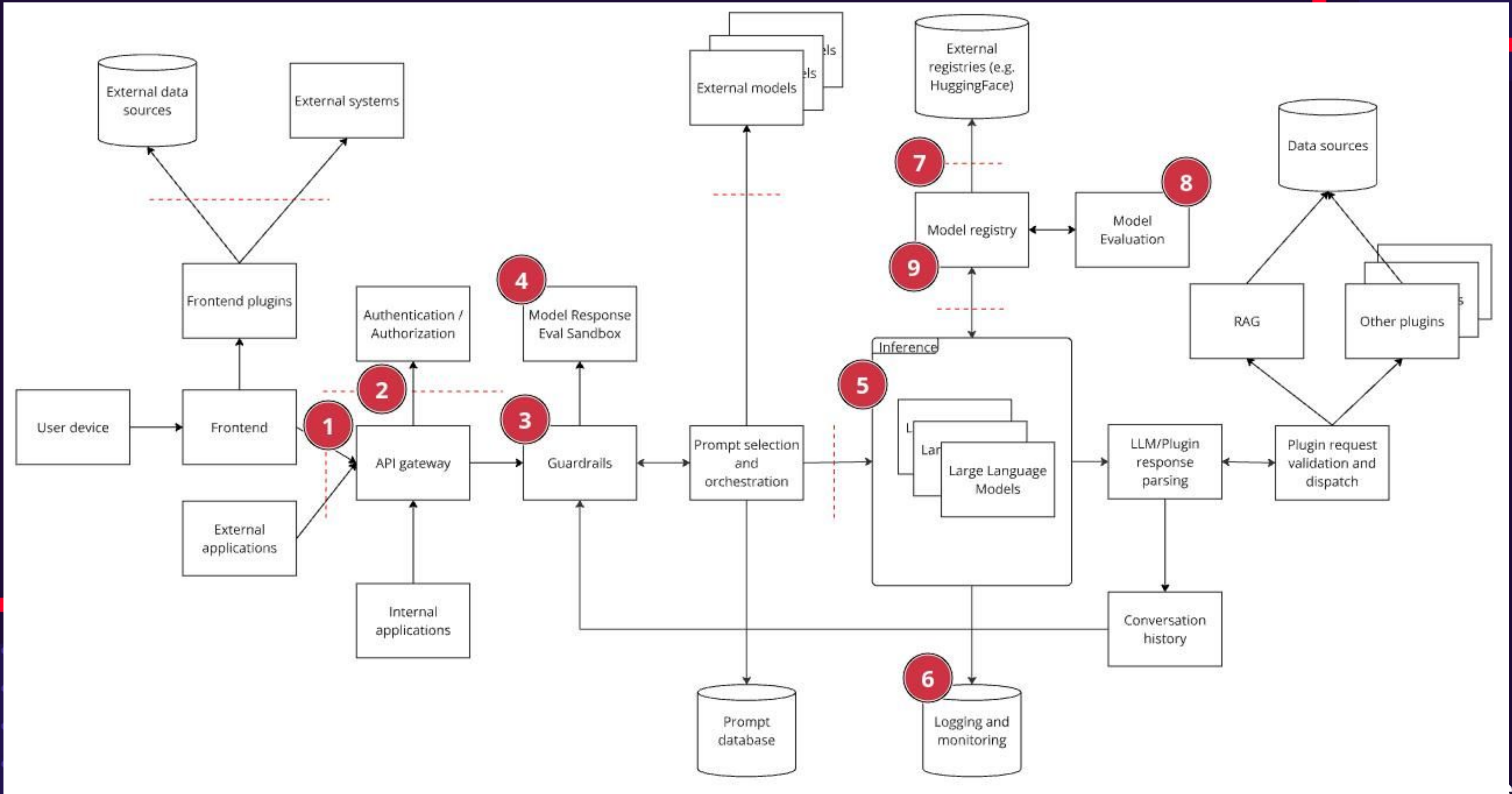
NIST AI RMF

Safe
Secure & Resilient
Explainable & Interpretable
Privacy-Enhanced
Fair - With Harmful Bias Managed
Accountable & Transparent
Valid & Reliable

OWASP Top 10 for LLMs

- LLM01:2025 Prompt Injection
- LLM02:2025 Sensitive Information Disclosure
- LLM03:2025 Supply Chain
- LLM04:2025 Data and Model Poisoning
- LLM05:2025 Improper Output Handling
- LLM06:2025 Excessive Agency
- LLM07:2025 System Prompt Leakage
- LLM08:2025 Vector and Embedding Weaknesses
- LLM09:2025 Misinformation
- LLM10:2025 Unbounded Consumption

AI application architecture



The need to have security at the platform

- Authentication and authorization
- Workload isolation
- Network security
- Continuous vulnerability scanning and analysis (Red Hat Advanced Cluster Security for Kubernetes)
- Immutable infrastructure
- Policy-driven configuration management
- Least privilege enforcement

Considerations for enterprises

- **Adopt** defense-in-depth strategy
- **Unify** AI security with enterprise cybersecurity
- **Holistic** approach by combining security, governance, compliance & AI-specific safety controls

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Next steps

Review AI Deployments

Review for security & safety gaps

Integrate AI Risk

Integrate AI risk into existing governance and security practices

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